The Toroidal Geometry of LLM Representations: From Detection to Karmonic-Guided Alignment

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Abstract

We present evidence that large language model hidden states exhibit toroidal topology and introduce a training-time regularizer that exploits this structure for truthfulness alignment. Using persistent homology on Qwen 2.5-0.5B hidden states extracted from TruthfulQA, we detect strong toroidal signals across all network layers: Betti numbers $\beta_1 = 23$–37 (indicating many independent cycles) and $\beta_2 \geq 1$ (indicating voids), with truthful and hallucinated answers separated by 30–121° in PCA angular space. Motivated by this geometric finding, we replace the generic Sinkhorn optimal transport regularizer in the ENIGMA alignment framework with **Karmonic spectral filtering**—a targeted regularizer derived from the cycle graph Laplacian that preserves low-frequency toroidal modes while attenuating high-frequency noise. In controlled ablation experiments using DPO preference optimization on TruthfulQA, Karmonic filtering outperforms Sinkhorn OT by **+1.3 percentage points** on MC1 accuracy and **+4.4 percentage points** on MC2 score in matched conditions (both with SAMI mutual information). DPO with Karmonic alone achieves MC1 = 0.468 on TruthfulQA, a **+19 percentage point** improvement over the untrained baseline, with stable perplexity. All experiments use a single RTX 4090 with 200 training steps.

Scope: This work is empirical. We detect toroidal topology and show that a torus-aware regularizer outperforms a geometry-agnostic one. We make no claims about universal structure or ontological status of the detected topology. The contribution is operational: Karmonic filtering is a drop-in replacement for Sinkhorn OT in geometry-aware alignment pipelines.

1 Introduction

Large language models (LLMs) routinely generate fluent but factually incorrect text—a phenomenon termed *hallucination* [9]. Mitigation strategies divide into inference-time methods (retrieval-augmented generation, logit manipulation, chain-of-thought prompting) and training-time methods (RLHF, DPO, constitutional AI). A recent line of work explores *geometric* approaches: if LLM representations occupy specific manifold structures, one can design interventions that respect or exploit that geometry.

Our prior work on toroidal logit bias (TLB) [10] demonstrated that inference-time toroidal constraints on token selection yield consistent truthfulness gains across four models (+0.2 to +2.8 percentage points on TruthfulQA). However, TLB operates post-hoc on logits without modifying the model's internal representations.

The ENIGMA framework [1] introduced a three-layer geometry-aware training pipeline combining group-relative policy optimization (GRPO) [3], symmetric alignment via mutual information (SAMI) [2], and Sinkhorn optimal transport (OT) regularization. On Gemma-3-1B, ENIGMA achieved +12 percentage points on TruthfulQA. However, its OT component is geometry-agnostic—it constrains drift in Wasserstein space without knowledge of the representation manifold's structure.

This paper bridges the gap between inference-time geometric intervention and geometry-agnostic training regularization. We make four contributions:

1. **Topological detection**: We show that LLM hidden states exhibit toroidal topology via persistent homology ($\beta_1 = 23$–37 across layers), providing empirical justification for torus-aware methods.
2. **Angular separation**: Truthful and hallucinated answers separate by 30–121° in PCA angular projections, confirming that the detected toroidal structure encodes task-relevant information.
3. **Karmonic spectral filtering**: We introduce a targeted replacement for Sinkhorn OT that exploits the cycle graph Laplacian eigenstructure, preserving low-frequency class structure while attenuating high-frequency noise.
4. **Empirical validation**: In controlled ablations, Karmonic beats Sinkhorn OT by +1.3pp MC1 and +4.4pp MC2, establishing that topology-aware regularization outperforms geometry-agnostic drift control.

2 Background

2.1 Toroidal Topology and the Tonnetz

The Tonnetz, from 19th-century music theory [13], arranges pitch classes on a torus $T^2 = S^1 \times S^1$ such that harmonically related tones are geometrically proximate. The discrete torus $C_N \times C_M$ (product of cycle graphs) inherits this property: toroidal distance captures relational proximity with wraparound connectivity and no boundary effects.

The relevance to language models is the hypothesis—tested empirically in this paper—that token and concept representations in LLMs inhabit manifolds with similar cyclic structure, where semantic proximity corresponds to angular proximity on toroidal coordinates.

2.2 Persistent Homology

Persistent homology [5, 6] computes topological features of a point cloud at multiple scales. The Vietoris-Rips complex $\mathrm{VR}_\epsilon(X)$ connects points within distance ϵ; as ϵ increases, topological features (connected components, cycles, voids) appear and disappear. Features persisting over large ϵ-ranges indicate genuine topology rather than noise.

Betti numbers β_k count independent k-dimensional features: β_0 counts connected components, β_1 counts independent cycles (loops), and β_2 counts enclosed voids. A torus T^2 has $(\beta_0, \beta_1, \beta_2) = (1, 2, 1)$. Finding $\beta_1 \gg 2$ in data suggests a high-dimensional torus T^k with $k > 2$.

2.3 Karmonic Spectral Filtering

The normalized Laplacian of the cycle graph C_N has eigenvalues:

$$\lambda_n = 2 - 2\cos\left(\frac{2\pi n}{N}\right), \quad n = 0, 1, \ldots, N-1 \tag{1}$$

The spectral gap $\lambda_1 = 2 - 2\cos(2\pi/N)$ governs mixing time and information propagation on the cycle. The Karmonic spectral filter [11] assigns mode-dependent weights to a Fourier basis on T^2: low-frequency modes (near λ_1) carry class-level structure and are preserved; high-frequency modes (near $\lambda_{\max}$) carry noise and are attenuated. This yields a *mode-weighted uniformity* loss:

$$\mathcal{L}_{\text{Karmonic}} = \sum_{n=1}^{M} w_n \cdot \mathrm{LogSumExp}\big(-t \cdot D_n^2\big) \tag{2}$$

where D_n^2 is the pairwise squared distance matrix of the n-th Fourier mode embedding, t is a temperature parameter, and $w_n = (\lambda_n - \lambda_1)/(\lambda_M - \lambda_1)$ weights each mode by its spectral position.

2.4 The ENIGMA Framework

ENIGMA [1] combines three geometric training objectives:

- **GRPO** [3]: Group-relative policy optimization on the Fisher-Rao manifold, where G completions per prompt are scored and advantages are computed relative to the group.
- **SAMI** [2]: Symmetric Alignment via Mutual Information. An InfoNCE objective maximizing $I(Y; C \mid X)$—mutual information between completion representations Y and constitutional principle embeddings C, conditioned on prompts X.
- **Sinkhorn OT**: Entropic optimal transport between current and reference hidden state distributions, constraining representation drift in Wasserstein space with entropic regularization parameter ϵ.

The unified loss is $\mathcal{L}_{\text{ENIGMA}} = \mathcal{L}_{\text{GRPO}} + \lambda_{\text{SAMI}}\mathcal{L}_{\text{SAMI}} + \lambda_{\text{OT}}R_{\text{OT}}$.

Our key observation: ENIGMA's OT component is *geometry-agnostic*—it constrains drift without exploiting the manifold's specific structure. If the representation manifold is toroidal (as we show), a *targeted* regularizer that knows the torus geometry should outperform generic drift control.

3 Detecting Toroidal Structure in LLM Hidden States

3.1 Method

We extract hidden states from Qwen 2.5-0.5B-Instruct [12] on the TruthfulQA validation set (817 questions). For each question, we process both the correct and incorrect MC1 answer choices through the model and capture hidden states at five layers: 0 (embedding), 12 (middle), 21, 22, and 23 (final). Each hidden state vector has dimension 896.

For each layer, we apply three independent tests:

1. Persistent homology (Ripser [6]): We subsample 500 points, apply PCA to 50 dimensions, and compute Vietoris-Rips persistence diagrams up to dimension 2. Significant features are those with persistence exceeding median $+ 1\sigma$.

2. Spectral analysis: We build a k-nearest-neighbor graph ($k = 10$) on 1000 subsampled points (PCA to 50 dimensions), compute the graph Laplacian $L = D - A$, and analyze the smallest eigenvalues for torus-consistent patterns (multiplicity-2 spectral gap).

3. Circular statistics: We project to multiple 2D PCA planes, convert to polar coordinates, test for non-uniform angular distribution (Rayleigh test, $p < 0.01$), and measure the angular separation between truthful and hallucinated answer centroids.

3.2 Results

3.3 Interpretation

Three findings emerge:

High-dimensional torus. All layers show $\beta_1 \gg 2$, ranging from 23 to 37 significant 1-cycles. This far exceeds the $\beta_1 = 2$ of a standard 2-torus, suggesting the hidden states live on or near a high-dimensional torus T^k with $k \gg 2$. The middle layer (12) shows the richest topology ($\beta_1 = 37$, $\beta_2 = 21$), consistent with findings that middle layers encode the most task-relevant features.

Layer	Score	β_1	β_2	Best Angular Sep.	Best Plane
0 (embedding)	6/7	32	8	120.7°	PC2–PC3
12 (middle)	6/7	37	21	35.8°	PC1–PC2
21	6/7	23	2	30.2°	PC2–PC4
22	5/7	26	3	30.0°	PC2–PC3
23 (final)	6/7	25	4	35.0°	PC2–PC3

Table 1: Persistent homology and circular statistics across layers of Qwen 2.5-0.5B on TruthfulQA. β_1 counts significant 1-cycles (persistence $>$ median $+ 1\sigma$). Score combines persistent homology (0–3), spectral multiplicity (0–1), and circular statistics (0–3).

Truth–hallucination angular separation. Truthful and hallucinated answers separate by 30–121° in PCA angular space, with the embedding layer showing the largest separation (120.7°). This confirms that the toroidal structure encodes task-relevant information—truthful and hallucinated representations occupy geometrically distinct toroidal regions.

Learned projection needed. A random projection to T^2 (via randomly initialized FourierTorusHead) yields near-zero separation ($< 1°$), confirming that the truthfulness-relevant toroidal dimensions must be *learned*. This motivates training a FourierTorusHead jointly with a content signal (DPO/SAMI) to discover the task-relevant toroidal coordinates.

4 Karmonic-Guided Alignment

4.1 Architecture

We replace ENIGMA's Sinkhorn OT with Karmonic spectral filtering while retaining the preference optimization and (optionally) mutual information components. Our architecture:

Listing 1: Karmonic alignment architecture (simplified)

```
class FourierTorusHead(nn.Module):
    """Projects hidden states to torus coordinates."""
    def __init__(self, input_dim=896, hidden=128,
                 torus_dim=2, n_modes=6):
        self.net = nn.Sequential(
            nn.Linear(input_dim, hidden),
            nn.LayerNorm(hidden), nn.ReLU(),
            nn.Linear(hidden, torus_dim))
        self.modes = torch.arange(1, n_modes + 1)

    def forward(self, z):
        angles = 2*pi * sigmoid(self.net(z))
        # Fourier embedding: cos/sin at each mode
        n_angles = angles.unsqueeze(-1) * self.modes
        fourier = stack([cos(n_angles), sin(n_angles)])
        return angles, fourier.reshape(B, -1)
```

The full training pipeline:

1. **DPO preference loss**: Direct preference optimization [4] between known correct and incorrect TruthfulQA answers: $\mathcal{L}_{\text{DPO}} = -\log \sigma\big(\beta \cdot (\log \pi_\theta(y_w \mid x) - \log \pi_\theta(y_l \mid x))\big)$, where y_w and y_l are the preferred and dispreferred answers.

2. **SAMI mutual information** (optional): Symmetric InfoNCE maximizing $I(Y; C \mid X)$ between completion hidden states and constitutional principle embeddings.

3. **Karmonic filter loss**: Hook the final hidden layer, mean-pool across the sequence, apply gradient scaling ($\times 0.1$) to prevent the auxiliary head from dominating, project through FourierTorusHead, compute mode-weighted uniformity.

The Karmonic component uses a buffer mechanism to accumulate hidden states across 8 steps (since batch size is 1 for DPO), then computes the loss on the concatenated batch with only the current step's hidden states carrying gradients.

4.2 Why DPO Instead of GRPO

At the 0.5B model scale with single-GPU training, GRPO's on-policy generation produces nearly identical completions (low diversity), yielding zero group-relative advantages and zero gradient signal. DPO avoids this by directly comparing known correct and incorrect answers from TruthfulQA's MC1 annotations, providing a strong and consistent preference signal regardless of model scale.

4.3 Why Karmonic Should Beat Sinkhorn OT

Sinkhorn OT constrains the Wasserstein distance between current and reference hidden state distributions. This prevents catastrophic drift but is agnostic to the manifold's structure—it penalizes all movement equally, regardless of whether it moves along meaningful toroidal coordinates or noise dimensions.

Karmonic filtering exploits the specific eigenstructure of the torus:

- **Spectral selectivity**: Low-frequency modes (λ_n near λ_1) encode class-level structure and receive low penalty weight. High-frequency modes (λ_n near $\lambda_{\max}$) encode noise and receive high penalty.
- **Manifold-aware**: The penalty structure is derived from the cycle graph Laplacian, matching the detected toroidal topology.
- **Computational efficiency**: No iterative Sinkhorn solve; the loss is a direct LogSumExp over pairwise distances per mode.

4.4 Experimental Setup

We test five conditions in controlled ablation:

#	Condition	DPO	SAMI	Regularizer
1	DPO only	✓		—
2	DPO + SAMI	✓	✓	—
3	DPO + SAMI + OT	✓	✓	Sinkhorn
4	DPO + SAMI + Karmonic	✓	✓	Karmonic
5	DPO + Karmonic	✓		Karmonic

Table 2: Five experimental conditions. All use Qwen 2.5-0.5B-Instruct with LoRA ($r = 16$, $\alpha = 32$).

Hyperparameters: LoRA targets $\{q, k, v, o\}$ projections with dropout 0.1. DPO $\beta = 0.1$. Learning rate 2×10^{-5} with AdamW (weight decay 0.01). SAMI: $\lambda_{\text{SAMI}} = 0.05$, temperature 0.07, 6 constitutional principles, row InfoNCE (column skipped when batch $<$ num principles). Karmonic: $\lambda_{\text{Karmonic}} = 0.01$, gradient scale 0.1, grid size 12, 6 modes, buffer size 8. Sinkhorn OT: $\lambda_{\text{OT}} = 0.01$, $\epsilon = 0.1$, 20 iterations. All conditions: 200 steps, 500 training samples from TruthfulQA, evaluated on the full 817-sample validation set.

Hardware: Single NVIDIA RTX 4090 (RunPod), ∼1.4 GB VRAM, ∼1 minute training + ∼6 minutes evaluation per condition.

5 Results

5.1 Main Results

Condition	MC1	MC2	PPL	ΔMC1	ΔMC2
Untrained baseline	0.278	0.407	18.06	—	—
DPO only	**0.486**	**0.554**	18.24	+20.8pp	+14.7pp
DPO + SAMI	0.416	0.492	18.18	+13.8pp	+8.6pp
DPO + SAMI + OT	0.448	0.499	18.18	+17.0pp	+9.3pp
DPO + SAMI + Karmonic	0.461	**0.544**	18.22	+18.4pp	+13.7pp
DPO + Karmonic	0.468	0.531	18.21	+19.0pp	+12.5pp

Table 3: Main results on TruthfulQA (817 samples). MC1 = single-best accuracy, MC2 = normalized probability mass on correct answers, PPL = WikiText-2 perplexity. Deltas computed vs. untrained baseline. All conditions: Qwen 2.5-0.5B, LoRA $r = 16$, 200 steps.

5.2 DPO is Highly Effective

DPO alone produces a massive +20.8 percentage point MC1 improvement (0.278 → 0.486), confirming that direct preference optimization between known correct and incorrect answers provides strong alignment signal even at the 0.5B scale and with only 200 training steps. This serves as a strong baseline against which to measure regularizer contributions.

5.3 Karmonic Outperforms Sinkhorn OT

The headline result: in matched SAMI conditions (conditions 3 vs. 4):

- **MC1**: Karmonic 0.461 vs. OT 0.448 (**+1.3pp**)
- **MC2**: Karmonic 0.544 vs. OT 0.499 (**+4.4pp**)

The MC2 improvement is particularly striking: Karmonic not only selects the single best answer more accurately but also assigns substantially more probability mass to the full set of correct answers. This is consistent with Karmonic's spectral filtering mechanism, which preserves the low-frequency class structure that maps truthful answers to a coherent toroidal neighborhood.

5.4 SAMI Interference with DPO

An unexpected finding: adding SAMI to DPO *hurts* performance (MC1 drops from 0.486 to 0.416, −7pp). This contrasts with ENIGMA's results where SAMI improved GRPO. The likely explanation: SAMI's symmetric InfoNCE loss competes with DPO for gradient signal. DPO directly optimizes answer preferences, while SAMI optimizes alignment between completions and constitutional principles—at batch size 1, these objectives may conflict. Notably, both regularizers (OT and Karmonic) partially recover the SAMI-induced degradation, with Karmonic recovering more (0.461 vs. 0.448).

5.5 Perplexity Stability

All conditions maintain perplexity within the narrow range 18.18–18.24, confirming that neither DPO training nor Karmonic regularization degrades general language modeling quality. This is expected given the small number of training steps (200) and low LoRA rank (16).

6 Discussion

6.1 Why Karmonic Beats OT

The mechanism is spectral selectivity. Sinkhorn OT computes a transport plan between hidden state distributions that minimizes total movement, but it treats all dimensions equally. In a high-dimensional space with toroidal structure, most dimensions encode noise—penalizing movement along them wastes gradient budget.

Karmonic filtering decomposes the representation into Fourier modes on the torus and applies differential penalties: low modes (carrying class structure) are lightly regularized, high modes (carrying noise) are strongly regularized. This allows the model to freely adjust its class-relevant toroidal coordinates (guided by DPO) while constraining noise accumulation.

The +4.4pp MC2 advantage over OT reflects this: by preserving the low-frequency mode structure, Karmonic allows the model to distribute probability mass across all correct answers in a coherent toroidal neighborhood, rather than concentrating on a single one.

6.2 Limitations

1. **Model scale**: All experiments use Qwen 2.5-0.5B (0.5 billion parameters). Behavior at 1B+ scale—where ENIGMA reports its results [1]—may differ. The toroidal structure detected in Phase 2 should be re-verified at larger scales.

2. **Training duration**: 200 steps is minimal. Longer training may change the relative ordering of conditions, particularly for SAMI which may need more steps to converge.

3. **Single benchmark**: We evaluate on TruthfulQA only. Performance on reasoning (GPQA), coding (HumanEval), or open-ended generation is untested.

4. **DPO vs. GRPO**: We use DPO instead of ENIGMA's GRPO due to scale constraints. The relative advantage of Karmonic over OT should be validated under GRPO at larger scale.

5. **Torus dimensionality**: We project to T^2 (2-torus) via FourierTorusHead, but the persistent homology suggests T^k with $k \gg 2$. Higher-dimensional torus projections may capture more structure.

6.3 Connection to Toroidal Logit Bias

Our prior work [10] showed that inference-time toroidal constraints on logits yield +0.2 to +2.8pp TruthfulQA gains. The present work shows that training-time toroidal regularization yields much larger gains (+19pp). Together, these suggest a "full-stack" toroidal approach: Karmonic-guided training to shape representations, followed by TLB at inference to refine token selection. Testing this combination is future work.

7 Conclusion

We have presented three results that together establish toroidal geometry as a productive framework for LLM alignment:

1. **Toroidal topology is present**: Persistent homology detects $\beta_1 = 23$–37 significant cycles across all layers of Qwen 2.5-0.5B, with truthful/hallucinated angular separation of 30–121°.

2. **Karmonic beats Sinkhorn OT**: As a drop-in replacement for ENIGMA's OT regularizer, Karmonic spectral filtering achieves +1.3pp MC1 and +4.4pp MC2 in matched conditions, by exploiting spectral selectivity on the detected toroidal structure.

3. **DPO + Karmonic is effective**: At 0.5B scale with 200 steps, DPO with Karmonic regularization achieves MC1 = 0.468 (+19pp over baseline) with stable perplexity.

The broader implication is that *topology-aware regularization outperforms geometry-agnostic drift control*. When the representation manifold has identifiable structure—as we show LLM hidden states do—regularizers that exploit that structure provide better alignment outcomes.

Future work: Scale to 1–3B models with GRPO; test Karmonic with higher-dimensional torus projections (T^k, $k > 2$); combine training-time Karmonic with inference-time TLB; verify post-training amplification of toroidal structure (β_1 increase).

Code and data available at `https://github.com/Paraxiom/topological-coherence`.

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